

Name: _____ Date: _____

Definition 1. Recall the *quotient rule* (“low d-high minus high d-low over low-low”):

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{g(x) \frac{df}{dx} - f(x) \frac{dg}{dx}}{g^2(x)} \quad \text{or} \quad \left(\frac{f(x)}{g(x)} \right)' = \frac{g(x)f'(x) - f(x)g'(x)}{g^2(x)}$$

Problem 1. For each of the following functions, use the quotient rule to find the derivative $\frac{dy}{dx}$. Hint: you may need also need to use the power rule or product rule at some point for some problems!

a) $y = \frac{x}{\sin(x)}$

b) $y = \frac{\ln(x)}{x}$

c) $y = \frac{\sin(x)}{\cos(x)}$

d) $y = \frac{x^2 + 2x - 1}{\tan(x)}$

e) $y = \frac{e^x}{\ln(x)}$

f) $y = \frac{\cos(x)}{x^2 + 2}$

g) $y = \frac{xe^x}{\sin(x)}$

$$i) \ y = \frac{\sin(x) \cos(x)}{\sqrt{x}}$$

$$j) \ y = \frac{\sqrt{x}}{e^x}$$

$$k) \ y = \frac{\sqrt{x}}{2x^3 - 3}$$

$$\ell) \ y = \frac{xe^x}{\ln(x)}$$

$$m) \ y = \frac{\sin(x)}{(x-4) \ln(x)}$$

$$n) \ y = \frac{\cos(x)}{xe^x}$$

$$o) \ y = \frac{\tan(x) \ln(x)}{e^x \sqrt{x}}$$